SESSION 06

Bank Branch Security Lab

Phase 6 · Detection completed: brute-force simulation and file integrity · Tuesday, October 13, 2026

Session goal	Clear the blocker left from last session, then finish the detection phase: run a brute-force attack end to end and add file integrity monitoring so Wazuh sees files change.
Outcome	**Detection phase complete.** Fixed the Guest Additions fault by mounting the ISO with Windows itself; ran a brute-force attack from BR01-WS01 that locked wade.wilson and triggered a Wazuh level-9 alert; added real-time file integrity monitoring on the Branch01 share, verified with add, modify and delete events.
Systems touched	BR01-WS01 (Guest Additions installed) · DC01 (Wazuh FIM configured) · WAZUH01 (alerts)
Branch network	No topology changes · 10.10.10.0/24
Skills shown	Hypervisor troubleshooting, ISO mounting, attack simulation, SIEM alert analysis, log correlation, file integrity monitoring, config management
Next session	Phase 7, the self-hosted AI assistant: a local model (Ollama) with an MCP tool for promotion matching, on synthetic data.

Executive summary

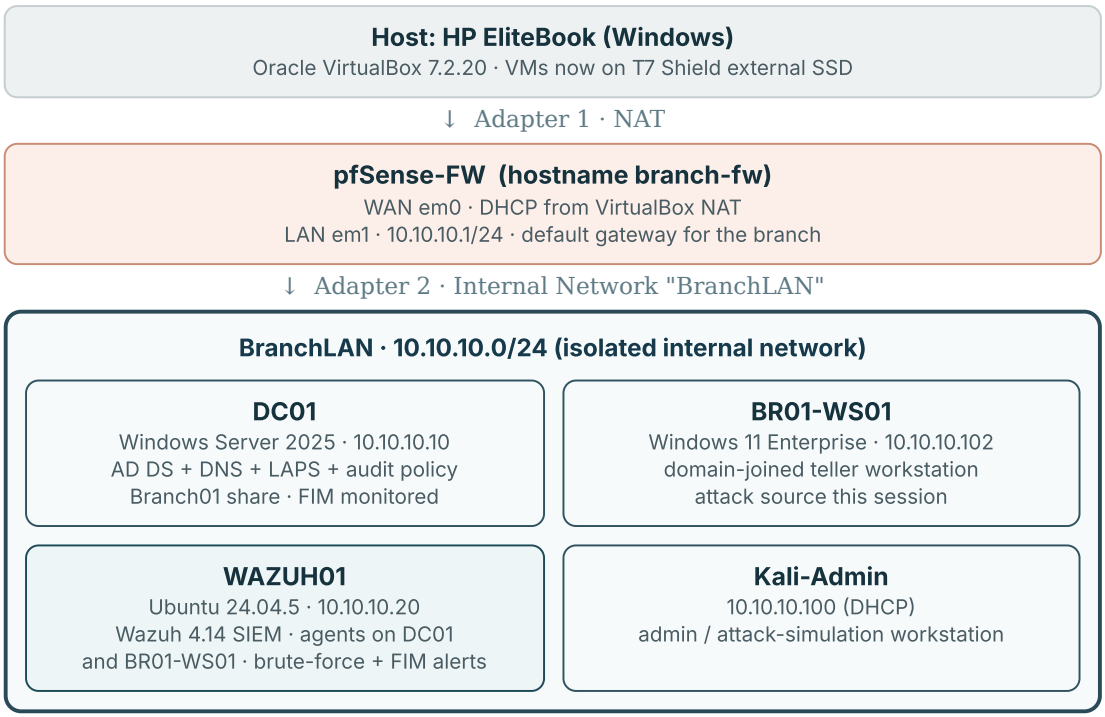
This session closed the two items hanging from last time and finished the detection phase. I resolved the **Guest Additions** fault by bypassing the wedged virtual CD and mounting the installer ISO with Windows itself, which restored the clipboard. I then ran the **brute-force simulation** from BR01-WS01 against wade.wilson on the domain controller; the lockout policy locked the account after five tries, and Wazuh escalated the burst to a high-severity alert, rule 60115 at level 9, which also named the source machine. Finally I added **file integrity monitoring** on the Branch01 share and proved it by creating, modifying and deleting a test file, each producing its own alert in seconds. The secure branch is now built, hardened, and monitored end to end.

Objectives and status

Objective	Status
Fix the Guest Additions / optical-drive blocker on BR01-WS01	☑ Complete
Run a brute-force attack end to end against the domain controller	☑ Complete
Confirm Wazuh escalates the attack to a high-severity alert	☑ Complete
Add real-time file integrity monitoring on the Branch01 share	☑ Complete
Verify FIM catches file add, modify and delete	☑ Complete
Snapshot all five VMs (Post-Session06)	☐ To do at session end

1. Network design

No topology changes this session. The work was on two existing machines: BR01-WS01 (the attack source and the Guest Additions fix) and DC01 (the attack target and the file integrity monitor). The current branch looks like this.



Attack path exercised this session

Stage	What happened
Source	BR01-WS01 (a domain-joined workstation), standing in for a compromised branch PC
Target	wade.wilson on DC01 (10.10.10.10) over SMB/WMI
Control that fired	Domain lockout policy from Session 04 locked the account after five bad attempts
Detection	Wazuh rule 60115 "User account locked out (multiple logon failures)" at level 9, on agent DC01

2. Lab environment

Guest Additions, installed on BR01-WS01

Why it was needed	Shared clipboard and drag-and-drop depend on it; without it, paste into the VM did nothing
VirtualBox version	7.2.20 (confirmed from the mounted ISO label VBox_GAs_7.2.20)
Installer source	`VBoxGuestAdditions_7.2.20.iso`, downloaded inside the VM from download.virtualbox.org
How it was mounted	Windows-native mount via `Mount-DiskImage`, bypassing the broken VirtualBox virtual CD
Result	`VBoxService` running; clipboard and paste restored

File integrity monitoring, configured on DC01

Agent config file	`C:\Program Files (x86)\ossec-agent\ossec.conf` (backed up to `.bak` first)
Monitored path	`C:\Shares\Branch01` (the Tellers / Managers / IT / Common share), recursively
Mode	Real time (`realtime="yes"`) — alerts the instant a file changes
Depth of check	`check_all="yes"` — size, permissions, owner and file hashes
Content diffs	`report_changes="yes"` — records what changed inside text files
Service	Wazuh Windows agent, service name `WazuhSvc`; restarted to load the config

The FIM line added to ossec.conf

```
<syscheck>
    <directories check_all="yes" realtime="yes" report_changes="yes">C:\Shares\Branch01</directories>
    ...
</syscheck>
```

3. Work performed

Part A · Fixing the Guest Additions blocker

1. **Confirmed the fault persisted**: after a full VirtualBox restart, `Get-CimInstance Win32_CDROMDrive` still showed `MediaLoaded: False` on BR01-WS01's virtual CD.
2. **Decided to bypass the virtual CD** rather than keep fighting it, since Windows can mount an ISO with its own driver, independent of VirtualBox.
3. **Found the VirtualBox version** (7.2.20) and downloaded the matching `VBoxGuestAdditions_7.2.20.iso` inside BR01-WS01 using Edge.
4. **Right-click Mount failed** with "You don't have permission to mount the file." Checked the sparse flag with `fsutil sparse queryflag` — the file was not sparse, so it was an Explorer-level block, not a real permission issue.
5. **Mounted it from PowerShell** with `Mount-DiskImage`, which succeeded where Explorer would not. The ISO came up as drive E:.
6. **Installed Guest Additions** by running `E:\VBoxWindowsAdditions-amd64.exe`, then rebooted.
7. **Verified** with `Get-Service VBoxService` (Running). Clipboard and paste worked again, closing a problem that had spanned two sessions.

Part B · Brute-force simulation

8. **Reset the target**: `Unlock-ADAccount wade.wilson` on DC01 so the test started from a clean state.
9. **Ran the attack from BR01-WS01**: a PowerShell loop tried ten wrong passwords for `JRBANKNA\wade.wilson` against DC01 over WMI. All ten failed, as intended.
10. **The lockout policy fired**, locking wade.wilson after the fifth attempt, exactly as configured in Session 04.
11. **Wazuh escalated it**: rule **60115** "User account locked out (multiple logon failures)" at **level 9** on agent DC01 — distinct from the level-5 single-failure alert.
12. **Confirmed the detail** in the alert: `targetUserName: wade.wilson`, event **4740**, and `targetDomainName: BR01-WS01`, which on a 4740 event carries the caller computer, naming the source of the attack.

Part C · File integrity monitoring

13. **Located the share** with `Get-SmbShare`: Branch01 at `C:\Shares\Branch01`.
14. **Added a real-time FIM rule** for that path to DC01's `ossec.conf` (with a backup), then restarted `WazuhSvc`.
15. **Tested the full lifecycle**: created, modified and deleted a test file in the share.
16. **Verified three alerts** in Wazuh on DC01: rule 554 "File added" (level 5), rule 550 "Integrity checksum changed" (level 7), and rule 553 "File deleted" (level 7).

4. Design decisions and rationale

These are the choices made this session and the reasoning to give in an interview.

Decision	Rationale
Bypass the virtual CD instead of fixing it	After five attempts across two sessions, the VirtualBox optical device was clearly wedged. Windows' own ISO mounting is a separate mechanism, so routing around the broken part was faster and more reliable than continuing to repair it.
Match Guest Additions to VirtualBox 7.2.20	A version mismatch can install but may warn or misbehave. Matching avoids a whole class of problems.
Mount with PowerShell, not Explorer	Right-click Mount threw a misleading "permission" error. `Mount-DiskImage` uses the same Windows mounting service without the Explorer shell restriction, and it worked.
Attack from BR01-WS01, not Kali	Windows Server 2025 uses SMB 3.1.1 with mandatory signing, which the Kali tools could not negotiate. A domain-joined workstation speaks the right dialect natively, and a compromised internal PC is a more realistic scenario anyway.
Real-time FIM	Scheduled scans miss fast activity. Real time alerts the instant a file changes, which is what catches ransomware before it finishes.
check_all and report_changes	Hashing every file catches silent content edits; reporting changes shows what text changed, which turns an alert into evidence.
Monitor the whole share recursively	Sensitive files live across the Tellers, Managers, IT and Common folders, so watching the share root covers all of them in one rule.
Back up ossec.conf before editing	A broken config stops the agent reporting. A `.bak` copy makes any mistake a one-command rollback.

5. Troubleshooting log

Incident 1: virtual CD still would not load media

Symptom	After a full VirtualBox restart, BR01-WS01's optical drive still reported `MediaLoaded: False`.
Root cause	The VM's virtual optical device was wedged in a way a restart did not clear — a VirtualBox-level fault, not a Windows one.
Resolution	Stopped repairing the device. Downloaded the Guest Additions ISO inside the VM and mounted it with Windows' own driver, which does not touch the VirtualBox optical device at all.
Lesson	When a component stays broken after every reasonable fix, route around it. The goal was Guest Additions, not a working virtual CD.

Incident 2: "You don't have permission to mount the file"

Symptom	Right-click › Mount on the downloaded ISO failed with a permission error, even as an administrator.
Root cause	Not an account-permission problem. Explorer refuses to mount some freshly downloaded ISOs; the usual culprits are a sparse-file flag or the internet "mark of the web." A sparse check came back negative, so it was an Explorer-shell block.
Resolution	Mounted it from PowerShell with `Mount-DiskImage`, which uses the same Windows mounting service without the shell restriction. It attached immediately as drive E:.
Lesson	A "permission" message is not always about your account. When the GUI blocks an action, the command line often does the same thing without the extra gatekeeping.

Incident 3 (carried from Session 05, now closed): Kali could not brute-force the DC

Symptom	crackmapexec, netexec and hydra all fingerprinted DC01 but timed out on every authentication attempt.
Root cause	Windows Server 2025 negotiates SMB 3.1.1 with mandatory signing; those Kali tools did not complete the handshake against that build.
Resolution	Ran the attack from BR01-WS01 over native Windows authentication instead. It worked first try and produced the lockout and the Wazuh alert.
Lesson	When several tools fail the same way, suspect the protocol, not the tools, and pick a vantage point that speaks the target's language.

6. Verification checklist

☑ = verified this session. ☐ = still to confirm; tick it by hand when done.

Check	How to verify	Result
☑ Guest Additions installed	`Get-Service VBoxService`	Pass · Running
☑ Clipboard / paste works in VM	Paste into PowerShell	Pass
☑ Brute-force attempts all failed	Attack script output	Pass · 10 of 10 failed
☑ Account locked by policy	Event 4740 on DC01	Pass · wade.wilson locked
☑ Wazuh high-severity alert	Threat Hunting, rule 60115	Pass · level 9
☑ Alert names the attack source	Alert field targetDomainName	Pass · BR01-WS01
☑ FIM rule applied	`WazuhSvc` restarted clean	Pass
☑ FIM catches a new file	Rule 554 "File added"	Pass · level 5
☑ FIM catches a change	Rule 550 "Integrity checksum changed"	Pass · level 7
☑ FIM catches a deletion	Rule 553 "File deleted"	Pass · level 7
☐ Post-Session06 snapshots (all 5 VMs)	Machine › Take Snapshot	To do at session end
☐ wade.wilson unlocked for normal use	`Unlock-ADAccount wade.wilson`	Do before next login test